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# Surgical, Radiologic, and Endoscopic Anatomy of the Kidney and Ureter

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## Keywords

anatomy; endoscopy; kidneys; pelvicalyceal system; radiology; ureters

## Questions

1. The exact position of the kidney within the retroperitoneum varies during:
    a. different phases of respiration.
    b. presence of anatomic anomalies.
    c. body position.
    d. a, b, and c
    e. a and c
2. Gerota fascia envelops the kidney and the adrenal gland on all aspects but remains open:
    a. inferiorly.
    b. laterally.
    c. medially.
    d. inferiorly and laterally.
    e. inferiorly and medially.

3. The white line of Toldt is the lateral reflection of posterior parietal peritoneum that covers:
    a. the ascending colon.
    b. the descending colon.
    c. the transverse colon.
    d. the ascending and descending colons.
    e. the ascending and transverse colon.
4. What are the columns of Bertin?
    a. Extensions of renal medulla between the pyramids
    b. Extensions of renal cortex between the pyramids
    c. Cortical extensions between renal lobules
    d. The collecting ducts
    e. None of the above
5. Compared with the liver, the normal adult kidneys in gray-scale ultrasound appear:
    a. hyperechoic.
    b. isoechoic.
    c. hypoechoic.
    d. b and c.
    e. variable, depending on the renal function.
6. Occlusion or injury to a segmental renal artery will cause:
    a. no pathologic conditions.
    b. opening of the collateral circulation.
    c. segmental renal infarction.
    d. an effect that depends on the availability of collaterals.
    e. renal atrophy.
7. Ureteropelvic junction obstruction may be commonly caused by:
    a. the lower anterior segmental artery when it passes anterior to the ureter.
    b. crossing of the ureter by any of the renal segmental arteries.
    c. the posterior segmental artery when it passes posterior to the renal pelvis.
    d. the posterior segmental artery when it passes anterior to the ureter.
    e. the lower anterior segmental artery when it passes posterior to the ureter.

8. Occlusion of a segmental renal vein results in:
    a. segmental renal congestion.
    b. segmental renal atrophy.
    c. no pathologic conditions.
    d. an effect that depends on the availability of collaterals.
    e. gross hematuria.
9. The main renal vasculature can be accurately identified with 100% sensitivity by:
    a. Doppler ultrasonography.
    b. computed tomography angiography (CTA).
    c. intravenous urography.
    d. noncontrast computed tomography.
    e. all of the above.
10. Medial displacement of both pelvic ureteral segments might result from:
    a. pelvic lipomatosis.
    b. postabdominoperineal surgery.
    c. retroperitoneal fibrosis.
    d. all of the above.
    e. a and c.
11. If the small ureteral arteries that anastomose in the ureteral adventitia are disrupted, this may result in:
    a. ureteral ischemia.
    b. ureteral stricture.
    c. a and b.
    d. no impact on ureteral blood supply.
    e. gross hematuria.
12. The blood supply of the mid-ureter is mostly:
    a. anterior.
    b. posterior.
    c. medial.
    d. lateral.
    e. any of the above.
13. What is the Mercier bar?
    a. The intramural ureter
    b. The bladder trigone

c. The interureteral ridge
d. Intraureteral valves
e. None of the above

14. The higher the grade of the ureteral orifice:
    a. the greater its tendency to be laterally located.
    b. the lesser its tendency to reflux.
    c. the greater its tendency to be associated with ureterocele.
    d. the greater its tendency to reflux.
    e. none of the above.

## Answers

1. **d. a, b, and c.** The exact position of the kidney within the retroperitoneum varies **during different phases of respiration, body position, and presence of anatomic anomalies**. For example, the kidneys move inferiorly about 3 cm (one vertebral body) during inspiration and during the changing of body position from supine to the erect position.
2. a. Inferiorly. **Gerota fascia encasing the kidneys, adrenal glands, and abdominal ureters is closed superiorly and laterally and serves as an anatomic barrier to the spread of malignancy and a means of containing perinephric fluid collections. Superiorly, the Gerota fascia is continuous with the diaphragmatic fascia on the inferior surface of the diaphragm, whereas inferiorly, the anterior and posterior layers of Gerota fascia are loosely attached where perinephric fluid collections can track inferiorly into the pelvis without violating Gerota fascia.**
3. **d. The ascending and descending colons.** To access the kidneys transperitoneally, the colon needs to be mobilized from the white line of Toldt, which is the lateral reflection of posterior parietal peritoneum **over the ascending and descending colon.**
4. **b. Extensions of renal cortex between the pyramids.** The renal cortex is about 1 cm in thickness and covers the base of each renal pyramid peripherally and **extends downward between the individual pyramids to form the columns of Bertin.**

5. **b. Isoechoic.** In adults, the normal kidneys have smooth margins and are **isoechoic to the liver**. However, both renal cortices and pyramids are usually hypoechoic to the liver, spleen, and renal sinus. Compared with renal parenchyma, renal sinus appears hyperechoic because of the presence of hilar adipose tissue, blood vessels, and lymphatics.
6. **c. Segmental renal infarction. After entering the hilum, each artery divides into five segmental end arteries that do not anastomose significantly with other segmental arteries. Therefore, occlusion or injury to a segmental branch will cause segmental renal infarction.** Nevertheless, the area supplied by each segmental artery could be independently surgically resected.
7. **d. The posterior segmental artery when it passes anterior to the ureter.** The posterior segmental artery from the posterior division **passes posterior to the renal pelvis** while the others pass anterior to the renal pelvis. **If the posterior segmental branch passes anterior to the ureter, ureteropelvic junction obstruction may occur.**
8. **c. No pathologic conditions. The renal venous drainage correlates closely with the arterial supply with the exception that unlike the arterial supply, venous drainage has extensive collateral communication through the venous collars around minor calyceal infundibula.** Furthermore, the interlobular veins that drain the **post-glomerular capillaries also communicate freely with perinephric veins** through the subcapsular venous plexus of stellate veins. Therefore, occlusion of a segmental venous branch **has little effect on venous outflow.**
9. **b. Computed tomography angiography (CTA).** Doppler ultrasonography clearly identifies renal arteries at their origin from the abdominal aorta. However, the main renal artery is often difficult to identify at baseline ultrasonography. Therefore, **CTA is currently considered the gold standard to assess renal arteries** with 100% sensitivity for identification of renal arteries and veins.
10. **d. All of the above.** Medial displacement of **both pelvic ureteral segments might result from retroperitoneal fibrosis, pelvic lipomatosis, or postabdominoperineal surgery.** However, medial

displacement and concavity of **a single pelvic ureter** may result from enlarged hypogastric nodes, a bladder diverticulum, or aneurismal dilatation of the hypogastric artery. Nevertheless, this may be a **normal finding in adult females** if only the right ureter is affected because of the uterine tilt to the left.

11. **c. a and b.** The abdominal portion of the ureter is supplied mainly by arterial branches medially from the main renal artery or the aorta, which form a longitudinal anastomosis on the ureteral wall. Despite this anastomotic plexus, **ureteral ischemia is not uncommon** if these small and delicate ureteral branches are disrupted. **Unnecessary lateral retraction and removal of the periureteral adventitial tissues containing the blood supply can result in ureteral ischemia and subsequent stricture.**
12. **b. Posterior.** Although main renal arteries or the aorta supply the abdominal ureter medially, the blood supply to the distal ureter comes laterally from the superior vesical artery, a branch of the internal iliac artery, and the mid-ureter is supplied by branches **arising posteriorly** from the common iliac arteries. **Therefore, the blood supply of the ureter is medial in the proximal part, posterior in the mid portion, and lateral in the distal portion.**
13. **c. The interureteral ridge.** Once the cystoscope is inside the bladder neck, the trigone can be seen as a raised, smooth triangle. The apex of that triangle is situated at the bladder neck; its base is formed by **the interureteral ridge or Mercier bar**, extending between the two ureteral orifices.
14. **a. The greater its tendency to be laterally located.** The ureteral orifices are classified according to their position or configuration. They are normally located at the medial aspect of the trigone (position A). However, they may be located at the lateral wall of the bladder or at its junction with the trigone (position C) or in between positions A and C (position B). In terms of configuration, grade 0 indicates a normal ureteral orifice that looks like a cone or a volcano. Grades 1, 2, and 3 describe stadium, horseshoe, and golf-hole orifice, respectively. **The higher the grade of the orifice, the greater its tendency to be laterally located and to reflux.**

## Chapter Review

1. The medial aspect of each kidney is rotated anteriorly 30 degrees.
2. The 12th rib overlies the right kidney; the 11th and 12th ribs overlie the left kidney.
3. The columns of Bertin contain the interlobar arteries.
4. Renal hilar structures from anterior to posterior are renal vein, renal artery, and renal pelvis.
5. The line of Brodel is an avascular plane between the anterior and posterior segments. It is variable in location and must be defined for each individual kidney.
6. Lumbar veins may drain directly into the renal veins, which occurs more commonly on the left. They may be the source of troublesome bleeding when dissecting around the renal vein.
7. Gerota fascia encasing the kidneys, adrenal glands, and abdominal ureters is closed superiorly and laterally and serves as an anatomic barrier to the spread of malignancy as well as a means of containing perinephric fluid collections. Superiorly, the Gerota fascia is continuous with the diaphragmatic fascia on the inferior surface of the diaphragm, whereas inferiorly, the anterior and posterior layers of Gerota fascia are loosely attached where perinephric fluid collections can track inferiorly into the pelvis without violating Gerota fascia.
8. Each renal artery divides into five segmental end arteries that do not anastomose significantly with other segmental arteries. They are end arteries and, when occluded, cause renal tissue ischemia and tissue atrophy.
9. The renal venous drainage has extensive collateral communication, and occlusion of a segmental vein will not impair the venous drainage to that segment.
10. Removal of the periureteral adventitial tissues containing the blood supply can result in ureteral ischemia and subsequent stricture.
11. The blood supply of the ureter is medially in the proximal part, posteriorly in the mid portion, and laterally in the distal portion.

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# Physiology and Pharmacology of the Renal Pelvis and Ureter

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## Keywords

contraction; obstruction; second messengers; sympathetic and parasympathetic nervous system; ureteral peristalsis

## Questions

1. During development, the ureteral lumen is obliterated and then recanalizes. Which of the following substances appears to be involved in this recanalization process?
    a. Prostaglandin $E_2$
    b. c-KIT
    c. Angiotensin
    d. Calcitonin gene-related peptide (CGRP)
    e. Acetylcholine
2. The resting membrane potential (RMP) is primarily determined by the distribution of which of the following ions across the cell membrane and the preferential permeability of the cell membrane to that ion?
    a. Potassium
    b. Sodium
    c. Calcium
    d. Chloride

e. Barium

3. With excitation of the ureteral muscle cell, an action potential is formed. Which of the following pairs of ions are primarily responsible for the upstroke of the action potential?
   a. Potassium and calcium
   b. Sodium and chloride
   c. Calcium and sodium
   d. Potassium and sodium
   e. Calcium and chloride
4. Which of the following must be phosphorylated for smooth muscle contraction to occur?
   a. Actin
   **b.Myosin**
   c. Calmodulin
   d. Calcium
   e. Troponin
5. The primary site for intracellular storage of calcium is:
   a. mitochondria.
   b. caveolae.
   c. the nucleolus.
   d. actin.
   e. the endoplasmic reticulum.
6. The second messenger involved in β-adrenergic agonist-induced ureteral relaxation is:
   a. cyclic adenosine monophosphate (cAMP).
   b. cyclic guanosine monophosphate (cGMP).
   c. nitric oxide.
   d. inositol 1,4,5-triphosphate ($IP_3$).
   e. diacylglycerol (DG).
7. The enzyme that degrades cyclic GMP is:
   a. guanylyl cyclase.
   b. myosin light-chain kinase.
   c. phosphodiesterase.
   d. phospholipase C.
   e. nitric oxide synthase (NOS).

8. The enzyme that degrades cyclic AMP is:
    a. adenylyl cyclase.
    b. myosin light-chain kinase.
    c. phosphodiesterase.
    d. phospholipase C.
    e. NOS.
9. Nitric oxide causes smooth muscle relaxation. In doing so, it activates which of the following enzymes?
    a. Guanylyl cyclase
    b. Myosin light-chain kinase
    c. Phosphodiesterase
    d. Phospholipase C
    e. NOS
10. The substrate for NOS is:
    a. cyclic AMP.
    b. cyclic GMP.
    c. GTP.
    d. **L-Arginine.**
    e. L-Citrulline.
11. Inducible NOS (iNOS) is:
    a. nicotinamide adenine dinucleotide phosphate (NADPH) independent and calcium independent.
    b. NADPH independent and calcium dependent.
    c. NADPH dependent and calcium independent.
    d. NADPH dependent and calcium dependent.
    e. nitric oxide dependent and calcium dependent.
12. The enzyme involved in the formation of DG (diacylglycerol) is:
    a. adenylyl cyclase.
    b. guanylyl cyclase.
    c. phosphodiesterase.
    d. protein kinase C.
    e. phospholipase C.
13. DG increases the activity of which enzyme?
    a. Adenylyl cyclase
    b. Guanylyl cyclase
    c. Phosphodiesterase

d. Protein kinase C
e. Phospholipase C

14. An agent that prevents reuptake of norepinephrine in nerve terminals and thus potentiates and prolongs the activity of norepinephrine is:
    a. tyrosine.
    b. monoamine oxidase.
    c. imipramine.
    d. tetramethylammonium.
    e. tetraethylammonium.
15. Norepinephrine is synthesized from:
    a. tyrosine.
    b. arginine.
    c. choline.
    d. cocaine.
    e. imipramine.
16. Which of the following inhibits ureteral and renal pelvic contractile activity?
    a. Substance P
    b. Neurokinin A
    c. Neuropeptide K
    d. Neuropeptide Y
    e. CGRP
17. The enzyme involved in prostaglandin synthesis is:
    a. phospholipase C.
    b. cyclooxygenase.
    c. protein kinase C.
    d. phosphodiesterase.
    e. adenosine triphosphate.
18. Which of the following is a β-adrenergic agonist?
    a. Cromakalim
    b. Physostigmine
    c. Propranolol
    d. Phenoxybenzamine
    e. Isoproterenol

19. Which of the following conditions must be present for urine to pass efficiently from the ureter into the bladder?
    a. Intraluminal ureteral contractile pressure must be above 40 cm $H_2O$.
    b. The ureterovesical junction (UVJ) must relax.
    c. Intraluminal ureteral contractile pressures must be greater than intravesical baseline pressures.
    d. Intravesical contractile pressures must be less than 40 cm $H_2O$.
    e. The bladder must relax just before contraction of the ureter.
20. What is normal baseline or resting ureteral pressure?
    a. 0 to 5 cm $H_2O$
    b. 5 to 10 cm $H_2O$
    c. 10 to 15 cm $H_2O$
    d. 15 to 20 cm $H_2O$
    e. 20 to 25 cm $H_2O$
21. The Laplace equation expresses the relationship between the variables that affect intraluminal pressure. Which of the following conforms to the Laplace relationship?
    a. Tension = (radius × wall thickness)/pressure
    b. Tension = (radius × pressure)/wall thickness
    c. Tension = (wall thickness × pressure)/radius
    d. Pressure = (radius × wall thickness)/tension
    e. Pressure = (radius × tension)/wall thickness
22. Factors that facilitate ureteral stone passage include:
    a. increased hydrostatic pressures proximal to the calculus and relaxation of the ureter in the region of the stone.
    b. increased hydrostatic pressures proximal to the calculus and contraction of the ureter in the region of the stone.
    c. decreased hydrostatic pressures proximal to the calculus and relaxation of the ureter in the region of the stone.
    d. decreased hydrostatic pressures proximal to the calculus and contraction of the ureter in the region of the stone.

e. decreased contractile pressures proximal to the calculus and contraction of the ureter in the region of the stone.

23. Which of the following hormones inhibits ureteral contractility?
    a. Bombesin
    b. Thyroxine
    c. Estrogen
    d. Aldosterone
    e. Progesterone
24. A drug that has efficacy in managing ureteral colic is:
    a. bethanechol.
    b. prostaglandin $F_{2\alpha}$.
    c. physostigmine.
    d. indomethacin.
    e. ephedrine.
25. Which of the following is a calcium-binding protein that plays a role in smooth muscle contraction?
    a. Connexin 43
    b. Calmodulin
    c. Cromakalim
    d. Survivin
    e. Myosin
26. In the ureter, the resting or the contractile force developed at any given length depends on the direction in which the change in length is occurring. This is referred to as:
    a. viscoelasticity.
    b. creep.
    c. hysteresis.
    d. stress relaxation.
    e. compensatory relaxation.
27. Which of the following is noted to be expressed before initiation of ureteral peristaltic activity?
    a. Prostanoids
    b. Nitric oxide
    c. c-KIT
    d. Myosin light chain

e. Phosphodiesterase

28. Ureteral pacemaker activity is amplified by:
    a. prostanoids.
    b. norepinephrine.
    c. CGRP.
    d. cyclic GMP.
    e. potassium channel openers.

## Answers

1. **c. Angiotensin.** At a point during development, the ureteral lumen is obliterated and then recanalizes. It appears that angiotensin, acting through the AT2 receptor, is involved in the recanalization process. Knockout mice for the ATR2 gene have congenital anomalies of the kidney and urinary tract, which include multicystic dysplastic kidneys, megaureters, and ureteropelvic junction (UPJ) obstructions.
2. **a. Potassium.** When a ureteral muscle cell is in a nonexcited or resting state, the electrical potential difference across the cell membrane, the transmembrane potential, is referred to as the *resting membrane potential.* The RMP is determined primarily by the distribution of potassium ions ($K^+$) across the cell membrane and by the permeability of the membrane to potassium ions.
3. **c. Calcium and sodium.** When the ureteral cell is excited, its membrane loses its preferential permeability to $K^+$ and becomes more permeable to calcium ions ($Ca^{2+}$) that move inward across the cell membrane, primarily through L-type $Ca^{2+}$ channels, and give rise to the upstroke of the action potential.
4. **b. Myosin.** The most widely accepted theory suggests that phosphorylation of myosin is involved in the contractile process.
5. **e. The endoplasmic reticulum.** Calcium release from tightly bound storage sites (i.e., the endoplasmic or sarcoplasmic reticulum) increases the $Ca^{2+}$ concentration in the sarcoplasm.
6. **a. Cyclic adenosine monophosphate (cAMP).** Cyclic AMP is believed to mediate the relaxing effects of β-adrenergic agonists in

a variety of smooth muscles.

7. **c. Phosphodiesterase.** Both cyclic AMP and cyclic GMP can cause smooth muscle relaxation. Cyclic GMP is synthesized from GTP by the enzyme guanylyl cyclase and is degraded to 5′-GMP by a phosphodiesterase.
8. **c. Phosphodiesterase.** Phosphodiesterases degrade both cyclic AMP and cyclic GMP. It has been demonstrated in the canine ureter that a variety of inhibitors can preferentially inhibit the breakdown of one or the other cyclic nucleotide.
9. **a. Guanylyl cyclase.** Nitric oxide released from the nerve activates the enzyme guanylyl cyclase in the smooth muscle cell, with the resultant conversion of guanosine triphosphate to cyclic GMP, with resultant smooth muscle relaxation.
10. **d. L-Arginine.** NOS converts L-arginine to nitric oxide and L-citrulline in a reaction that requires NADPH.
11. **c. NADPH dependent and calcium independent.** An inducible NOS isoform, iNOS, is NADPH dependent but $Ca^{2+}$ independent and has been identified in ureteral smooth muscle.
12. **e. Phospholipase C.** Some actions of $\alpha_1$-adrenergic and muscarinic cholinergic agonists and a number of other hormones, neurotransmitters, and biologic substances are associated with an increase in intracellular $Ca^{2+}$ and are related to changes in inositol lipid metabolism. These agonists combine with a receptor on the cell membrane, and the agonist-receptor complex, in turn, activates an enzyme, phospholipase C, that leads to the hydrolysis of polyphosphatidylinositol 4,5-bisphosphate, with the formation of two second messengers: $IP_3$ and DG.
13. **d. Protein kinase C.** DG binds to an enzyme, protein kinase C, causes its translocation to the cell membrane, and, by reducing the concentration of $Ca^{2+}$ required for protein kinase C activation, results in an increase in this enzyme's activity.
14. **c. Imipramine.** The greatest percentage of the norepinephrine is actively taken up (reuptake or neuronal uptake) into the neuron. Neuronal reuptake regulates the duration for which norepinephrine is in contact with the innervated tissue and thus regulates the

magnitude and duration of the catecholamine-induced response. Agents such as cocaine and imipramine (Tofranil, Mallinckrodt, Inc., Hazelwood, MO), which inhibit neuronal uptake, potentiate the physiologic response to norepinephrine.

15. **a. Tyrosine.** Norepinephrine, the chemical mediator responsible for adrenergic transmission, is synthesized in the neuron from tyrosine.
16. **e. CGRP.** Tachykinins and CGRP are neurotransmitters released from peripheral endings of sensory nerves. Tachykinins stimulate contractile activity, and CGRP inhibits contractile activity.
17. **b. Cyclooxygenase.** The "primary" prostaglandins, $PGE_1$, $PGE_2$, and $PGF_{2\alpha}$, are synthesized from the fatty acid arachidonic acid by enzymatic reactions involving two cyclooxygenase (COX) isoforms, COX-1 and COX-2.
18. **e. Isoproterenol.** Isoproterenol, a β-adrenergic agonist, depresses smooth muscle contractility.
19. **c. Intraluminal ureteral contractile pressures must be greater than intravesical baseline pressures.** The theoretical aspects of the mechanics of urine transport within the ureter were described in detail by Griffiths and Notschaele in 1983. [a] At normal flow rates, while the renal pelvis fills, a rise in renal pelvic pressure occurs, and urine is extruded into the upper ureter, which is initially in a collapsed state. The contraction wave originates in the most proximal portion of the ureter and moves the urine in front of it in a distal direction. The urine that had previously entered the ureter is formed into a bolus. **To propel the bolus of urine efficiently, the contraction wave must completely coapt the ureteral walls, and the pressure generated by this contraction wave provides the primary component of what is recorded by intraluminal pressure measurements.** The bolus that is pushed in front of the contraction wave lies almost entirely in a passive, noncontracting part of the ureter.
20. **a. 0 to 5 cm $H_2O$.** Baseline or resting ureteral pressure is approximately 0 to 5 cm $H_2O$.
21. **b. Tension = (radius × pressure)/wall thickness.** The Laplace equation expresses the relationship between the variables that

affect intraluminal pressure: pressure = (tension × wall thickness)/radius.

22. **a. Increased hydrostatic pressures proximal to the calculus and relaxation of the ureter in the region of the stone. Two factors that appear to be most useful in facilitating stone passage are an increase in hydrostatic pressure proximal to a calculus and relaxation of the ureter in the region of the stone.**
23. **e. Progesterone. Several studies have shown an inhibitory effect of progesterone on ureteral function. Progesterone has been noted to increase the degree of ureteral dilatation during pregnancy and to retard the rate of disappearance of hydroureter in postpartum women.**
24. **d. Indomethacin.** Indomethacin, by reducing pelviureteral pressure and thus pelviureteral wall tension, might eliminate some of the pain of renal colic that is dependent on distention of the upper urinary tract.
25. **b. Calmodulin.** With excitation, there is a transient increase in the sarcoplasmic $Ca^{2+}$ concentration from its steady-state concentration of $10^{-8}$ to $10^{-7}$ M to a concentration of $10^{-6}$ M or higher. At this higher concentration, $Ca^{2+}$ forms an active complex with the calcium-binding protein calmodulin. Calmodulin without $Ca^{2+}$ is inactive. The calcium-calmodulin complex activates a calmodulin-dependent enzyme, myosin light-chain kinase. The activated myosin light-chain kinase, in turn, catalyzes the phosphorylation of the 20,000-dalton light chain of myosin. Phosphorylation of the myosin light chain allows activation by actin of myosin $Mg^{2+}$-ATPase activity, leading to hydrolysis of ATP and the development of smooth muscle tension or shortening.
26. **c. Hysteresis.** Because the ureter is a viscoelastic structure, the resting or contractile force developed at any given length depends on the direction in which change in length is occurring and on the rate of length change. This is referred to as *hysteresis*; for the ureter, at any given length, the resting force is less and contractile force is greater when the ureter is allowed to shorten than when the ureter is being stretched.

27. **c. c-KIT.** This tyrosine kinase receptor is important in the development of pacemaker activity and peristalsis of the gut (Der-Silaphet et al., 1998). Pezzone and colleagues (2003) identified c-KIT-positive cells in the mouse ureter. The expression of c-KIT was noted to be upregulated in the embryonic murine ureter before its development of unidirectional peristaltic contractions (David et al., 2005). Incubation of isolated cultured embryonic murine ureters with antibodies that neutralize c-KIT activity alters ureteral morphology and inhibits unidirectional peristalsis. c-KIT-positive cells have been identified in the human ureter (Metzger et al., 2004).
28. **a. Prostanoids.** The ionic conduction underlying pacemaker activity in the upper urinary tract is due to the opening and slow closure of voltage-activated L-type $Ca^{2+}$ channels, which are amplified by prostanoids (Santicioli et al., 1995a). This is opposed by the opening and closure of voltage and $Ca^{2+}$-dependent $K^{+}$ channels. It has been suggested that prostaglandins and excitatory tachykinins, released from sensory nerves, help maintain autorhythmicity in the upper urinary tract through maintenance of $Ca^{2+}$ mobilization. Tetrodotoxin and blockers of the autonomic nervous system, both parasympathetic and sympathetic, have little effect on peristalsis, suggesting that autonomic neurotransmitters play little role in maintaining pyeloureteral motility.

## Chapter Review

1. Efficient propulsion of the urinary bolus depends on the ability of the walls of the ureter to coapt.
2. Autonomic neurotransmitters play little role in maintaining pyeloureteral motility even though the ureter is supplied by sympathetic and parasympathetic neurons.
3. Ureteral muscle fibers are arranged in a longitudinal, circumferential, and spiral configuration.
4. The ureter is a syncytial type of smooth muscle without discrete neuromuscular junctions.

5. Ureteral peristalsis can occur without innervation; however, the nervous system does play at least a modulating role in ureteral peristalsis, particularly the sympathetic nervous system.
6. α-adrenergic stimulation increases ureteral activity. β-adrenergic stimulation inhibits ureteral and renal pelvic activity.
7. Ureteral pressures can be as high as 20 to 80 cm of water during a contraction.
8. Pressure within the bladder during the storage phase is of paramount importance in determining the efficiency of urine transport across the UVJ.
9. Ureteral obstruction causes a gradual increase in ureteral length and diameter.
10. Infection impairs urine transport by reducing ureteral contractions; it also reduces compliance at the UVJ, which may permit reflux.
11. Progesterone has an inhibitory effect on ureteral function. Progesterone has been noted to increase the degree of ureteral dilatation during pregnancy and to retard the rate of disappearance of hydroureter in postpartum women. The obstruction of pregnancy is primarily due, however, to mechanical factors and secondarily due to the hormonal effects of progesterone.
12. Pacemaker cells have resting potentials less negative than non-pacemaker cells and are located near the pelvicalyceal border.
13. Phosphodiesterase inhibitors and α 1A antagonists cause ureteral smooth muscle relaxation.
14. UPJ obstruction may be anatomic or due to a disordered propagation of peristaltic activity. Some have suggested that the latter is due to an alteration in the configuration of the muscle fibers at the UPJ.
15. Ureteral decompensation will occur when there are sustained intravesicular pressures that exceed 40 cm $H_2O$.

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[a] Sources referenced can be found in *Campbell-Walsh urology, ed 11*, on the Expert Consult website.

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# Renal Physiology and Pathophysiology Including Renovascular Hypertension

*Thomas Chi, and Meyeon Park*

## Keywords

renal artery stenosis; renal pathophysiology; renal physiology

## Questions

1. Which of the following statements regarding erythropoiesis is FALSE?
    a. Reduced erythropoiesis and anemia are common in chronic renal disease.
    b. Erythropoiesis is inhibited by low circulating oxygen tension.
    c. During chronic inflammation, erythropoiesis is decreased.
    d. The kidney makes most of the erythropoietin in the body.
    e. There are erythropoietin receptors in many organs of the body.
2. Which of the following statements is TRUE about sodium and the kidney?
    a. By definition, hypernatremia is always associated with elevated total body sodium content.
    b. Normal compensation for hyponatremia is decreased antidiuretic hormone (ADH) secretion and thirst suppression.

c. Abnormal elevation of serum lipids can lead to a false, elevated measurement of serum sodium.
d. If asymptomatic hyponatremia does not improve within 24 hours, intravenous hypertonic saline should be started.
e. In therapy for symptomatic hyponatremia, the goal should be a normal serum sodium value of 135 mEq/L within 48 hours.

3. Which of the following statements regarding therapy for hyponatremia is FALSE?
   a. Fluid overload as a result of hypertonic saline infusion should be treated with a loop diuretic such as furosemide.
   b. Too-rapid correction can lead to a cerebral demyelination syndrome.
   c. Aggressive therapy should be discontinued when the serum sodium concentration is raised 10% or symptoms subside.
   d. Intranasal desmopressin is a useful adjuvant therapy.
   e. For acute severe hyponatremia with symptoms, a typical infusion rate of hypertonic saline would be 1 mL/kg/h.
4. Which of the following statements regarding potassium is FALSE?
   a. Angiotensin-converting enzyme (ACE) inhibitors may be a cause of hypokalemia.
   b. Potassium is primarily an intracellular ion.
   c. Acidosis drives potassium out of the cell into the circulation.
   d. High-sodium load in the distal tubule promotes potassium excretion.
   e. The upper limit for safe intravenous potassium infusion is 40 mEq/h.
5. Which of the following statements regarding hyperkalemia is FALSE?
   a. Hemolysis of the blood sample may falsely elevate the measured potassium.
   b. Hyperkalemia can cause peaked T waves on the electrocardiogram (ECG).
   c. All patients with a serum potassium value greater than 5.5 mEq/L require immediate therapy.
   d. Nebulized albuterol can reduce serum potassium by promoting an intracellular shift of potassium.

e. Intravenous calcium does not lower serum potassium but is given to protect the heart from the effects of hyperkalemia.

6. Which of the following statements regarding renal handling of acid is FALSE?
   a. Most bicarbonate is reabsorbed in the distal collecting tubule.
   b. Lungs can excrete volatile acid, but the kidneys must excrete fixed acid.
   c. Carbonic anhydrase catalyzes the production of H+ and $HCO^-_3$ from $H_2O$ and $CO_2$.
   d. Chronic respiratory acidosis should lead to increased H+ in the kidney.
   e. Ammonium ion (NH+4) is produced from glutamine, primarily by proximal tubular cells.
7. Which of the following statements regarding renal tubular acidosis (RTA) is FALSE?
   a. The hallmark of RTA type I is a hyperchloremic metabolic acidosis with a high urinary pH (>5.5) in the presence of persistently low serum $HCO^-_3$.b.Type I RTA is also called distal RTA.
   c. Type II RTA is more common in children.
   d. The hallmark of type IV RTA is hypokalemia.
   e. The form of RTA most commonly associated with renal calculi is type I.
8. Which of the following statements is TRUE about vitamin D metabolism?
   a. Vitamin D deficiency is uncommon in chronic renal failure.
   b. Dermally synthesized cholecalciferol is the most potent form of vitamin D.
   c. Dermally synthesized cholecalciferol must be hydroxylated by both the liver and kidney for maximal potency.
   d. Vitamin D activity is mediated through membrane-bound vitamin D receptors.
   e. Vitamin D increases renal excretion of calcium.
9. Which of the following statements regarding parathyroid hormone (PTH) is FALSE?

a. PTH secretion is increased by hypocalcemia.
b. PTH secretion is increased by hyperphosphatemia.
c. PTH receptors are found mainly in bone and kidney.
d. PTH increases calcium and phosphorus reabsorption in the distal tubule.
e. PTH helps regulate 1,25(OH)-vitamin D levels by increasing 1α-hydroxylase activity.

10. All of the following are important in GFR regulation EXCEPT:
    a. afferent arteriolar tone.
    b. distal tubule chloride concentrations.
    c. angiotensin II.
    d. nitric oxide.
    e. serum osmolality.
11. All of the following statements regarding GFR assessment are true EXCEPT:
    a. plasma creatinine is an accurate marker of early reductions in GFR.
    b. inulin clearance is an accurate but impractical measurement of GFR.
    c. twenty-four-hour creatinine clearance overestimates GFR by 10% to 20%.
    d. use of the four-variable modification of diet in renal disease (MDRD) formula improves the accuracy of the plasma creatinine.
    e. plasma urea is an unreliable estimate of GFR.
12. Which of the following statements about the proximal convoluted tubule is FALSE?
    a. It functions as a bulk transporter, rather than a fine-tuner of ultrafiltrate.
    b. It is able to increase or decrease reabsorption rates in response to changes in GFR.
    c. It has a minor role in sodium reabsorption.
    d. It reabsorbs 80% of filtered water, mainly through aquaporin-1 water channels.
    e. It is the major site of bicarbonate reabsorption.

13. All of the following statements are true regarding the loop of Henle EXCEPT:
    a. it is responsible for the generation of a hypertonic medullary interstitium, which is necessary for urinary concentration.
    b. it is able to increase or decrease reabsorption rates in response to changes in GFR.
    c. the descending limb is highly water permeable.
    d. the thin ascending limb actively reabsorbs sodium, chloride, and urea.
    e. the thick ascending limb is impermeable to water.
14. A 67-year-old male with stable renal function and a creatinine of approximately 3 mg/dL presents with peripheral vascular disease and a blood pressure of 160/70 mm Hg. He was a long-standing smoker but has recently stopped. He is referred for evaluation for treatment of "ischemic nephropathy." He has hyperlipidemia and had a myocardial infarction 3 years before presentation. His current medications are a calcium channel blocker, an adequate-dose diuretic, and a statin. Which of the following is the most appropriate next step?
    a. Refer the patient for magnetic resonance angiography (MRA) with gadolinium.
    b. Add an angiotensin-converting enzyme (ACE) inhibitor.
    c. Refer for an angiogram.
    d. Increase the diuretic.
    e. Observe for deterioration in renal function and then refer for an angiogram.
15. Which of the following is most similar to human renal vascular hypertension and is felt to be angiotensin dependent rather than volume dependent?
    a. The two-kidney, one-clip Goldblatt model
    b. The one-kidney, one-clip Goldblatt model
    c. The two-kidney, two-clip Goldblatt model
16. Since the results of the Coral trial have been published, which statement is true regarding progression to end-stage renal disease?
    a. Surgical intervention and/or percutaneous angioplasty with intervention is no longer ever indicated.

b. Medical intervention is superior to surgical intervention.
c. Medical intervention is superior to percutaneous angioplasty with stenting.
d. Medical intervention is equal to percutaneous angioplasty with stenting.
e. Percutaneous angioplasty with stenting is superior to medical intervention.

17. Which of the following statements is TRUE regarding surgical revascularization of a renal artery?
    a. A kidney less than 8 cm in length can be successfully revascularized.
    b. Retrograde filling of the distal renal artery by collateral circulation on radiographic or scintigraphic imaging studies is more likely to result in a successful surgical outcome.
    c. Patients who require renal vascular surgery do not have significant vascular disease elsewhere.
    d. Correction of a renal artery lesion and an aortic aneurysm need to be done simultaneously.
    e. A renal ostial lesion always requires surgical revascularization rather than percutaneous angioplasty and stenting.
18. Ischemic nephropathy results from:
    a. a reduction in renal blood flow and perfusion.
    b. proinflammatory cytokines and/or angiotensin II.
    c. an irreversible change in perfusion pressure.
    d. a failure of autoregulation alone.
    e. failure of development of collateral circulation.

## Answers

1. **b. Erythropoiesis is inhibited by low circulating oxygen tension.** Erythropoiesis is increased by low circulating oxygen tension.
2. **b. Normal compensation for hyponatremia is decreased antidiuretic hormone (ADH) secretion and thirst suppression.**
3. **d. Intranasal desmopressin is a useful adjuvant therapy.** Desmopressin is useful to treat hypernatremia caused by diabetes

insipidus.

4. **a. Angiotensin-converting enzyme (ACE) inhibitors may be a cause of hypokalemia.**
5. **c. All patients with a serum potassium value greater than 5.5 mEq/L require immediate therapy.** Patients with mild elevation of potassium, especially when chronic and not associated with ECG changes, do not require emergent therapy.
6. **a. Most bicarbonate is reabsorbed in the distal collecting tubule.** Most bicarbonate reabsorption in the kidney occurs in the proximal tubule.
7. **d. The hallmark of type IV RTA is hypokalemia.** RTA type IV is most commonly associated with hyperkalemia. Aldosterone deficiency or resistance leads to decreased secretion of potassium in the distal tubule.
8. **c. Dermally synthesized cholecalciferol must be hydroxylated by both the liver and kidney for maximal potency.** Cholecalciferol is minimally active, but potency increases 100 times after it is hydroxylated at the 1- and 25-position to form calcitriol.
9. **d. PTH increases calcium and phosphorus reabsorption in the distal tubule.** PTH increases phosphorus excretion in the kidney.
10. **e. Serum osmolality.** GFR is not affected significantly by serum osmolality.
11. **a. Plasma creatinine is an accurate marker of early reductions in GFR.**
12. **c. It has a minor role in sodium reabsorption.** The PCT accounts for 65% of sodium reabsorption, the most of any tubular segment.
13. **d. The thin ascending limb actively reabsorbs sodium, chloride, and urea.** Reabsorption of sodium, chloride, and urea occurs passively in the thin ascending limb.
14. **b. Add an angiotensin-converting enzyme (ACE) inhibitor.** According to the Coral trial, medical therapy consisted of an ACE inhibitor, a statin, and a diuretic. A calcium channel blocker was not necessarily part of the therapy unless blood pressure could not be controlled on an ACE inhibitor and a diuretic alone. The patient has significant renal failure, and gadolinium is relatively contraindicated because this may predispose to nephrogenic

systemic fibrosis. In fact, further investigation at this point is not indicated because the patient's renal function is stable. Similarly, an angiogram is not indicated. Increasing the diuretic is not indicated because the diuretic is already on board and the presumption is that the dose is adequate. The final choice is not indicated unless there is rapid deterioration of renal function, and even then, there are no data to support that intervening with angiography and percutaneous angioplasty is helpful.

15. **a. The two-kidney, one-clip Goldblatt model.** The other choices do not allow for natriuresis/diuresis from the opposite unclipped kidney. Thus, those two models are volume-dependent causes of hypertension rather than angiotensin dependent.
16. **d. Medical intervention is equal to percutaneous angioplasty with stenting.** The results of the trial are clearly outlined in the text and require no specific explanation.
17. **b. Retrograde filling of the distal renal artery by collateral circulation on radiographic or scintigraphic imaging studies is more likely to result in a successful surgical outcome.** Most patients with renal vascular disease have significant vascular disease elsewhere. Renal artery correction and aortic aneurysm correction need not be done simultaneously. A renal ostial lesion may be corrected by percutaneous angioplasty and stenting depending on its radiologic appearance.
18. **b. Proinflammatory cytokines and/or angiotensin II.** Renal blood flow and perfusion may sometimes be maintained at baseline in patients with ischemic nephropathy. Ischemic nephropathy can reverse, and it is not secondary to a failure of autoregulation alone, as explained in the text. Many patients with progressive ischemic nephropathy have collateral circulation.

## Chapter Review

1. GFR reflects total renal function.
2. GFR can be approximated by creatinine clearance.

3. Formulas based on patient's age, weight, and serum creatinine can best estimate GFR.
4. The nephron has different functional segments that control homeostasis.
5. Most resorption of bicarbonate and ions occurs in the proximal tubule.
6. The architecture of the loop of Henle allows a highly hypertonic interstitium to develop, which is crucial to maximal urinary concentration.
7. Sodium imbalances should be evaluated in the context of fluid status.
8. The best tools to determine the cause of a sodium disorder are the history, volume status, and urinary sodium.
9. Severe sodium deficit or excess must be corrected slowly.
10. Potassium is primarily intracellular.
11. Serum potassium levels reflect total body potassium, as well as the equilibrium between intra- and extracellular potassium.
12. Symptomatic hyperkalemia requires urgent intervention to prevent cardiac dysfunction.
13. Water movement dictates solute concentrations, especially important for sodium balance.
14. Regulation of water movement is highly dependent on antidiuretic hormone.
15. Insertion of aquaporin channels facilitates water reabsorption.
16. Physiologic chemical reactions require a narrow range of serum pH.
17. Acid is excreted through the lungs and the kidney.
18. Type 1 RTA (distal) is the only type associated with renal stones.
19. In acid-base disorders, first determine whether the kidney ($HCO^-_3$) or lungs ($pCO_2$) are responsible for the primary disorder, then determine whether the compensatory response is appropriate.
20. The renin-angiotensin system is one of multiple chemical mediators that act on renal vascular tone to control renal blood flow.
21. Endothelin is the most potent vasoconstrictor.
22. Nitric oxide and carbon monoxide are potent vasodilators.
23. Renal physiology influences bone metabolism through several hormonal actions.

24. In the kidney, vitamin D is hydroxylated into calcitriol which helps maintain systemic calcium and phosphate levels.
25. PTH influences the kidney to increase calcium reabsorption.
26. Renal artery stenosis is the most common cause of renovascular hypertension.
27. Atherosclerotic vascular disease accounts for 2/3 of cases of renal artery stenosis.
28. Four types of fibromuscular dysplasia can also cause renal artery stenosis. Of these types, medial fibroplasia is the most common and least likely to progress over time.
29. Suspect renovascular hypertension in patients with severe or abrupt-onset hypertension who otherwise have no risk factors for high blood pressure.
30. Two models of renovascular hypertension were established with classic dog model experiments—the two-kidney, one-clip and one-kidney, one-clip models.
31. Noninvasive diagnostic imaging, including MR angiography, CT angiography, and duplex ultrasonography, should be first-line screening to diagnosis renal artery stenosis.
32. Medical management for renovascular hypertension is the mainstay of therapy.
33. Percutaneous intervention to treat renal artery stenosis may be effective, but growing evidence calls into question its efficacy to control blood pressure or prevent renal failure long term.
34. While surgical treatment is no longer first-line therapy for renal artery stenosis, its role still exists for the appropriately selected patient.

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# Renal Insufficiency and Ischemic Nephropathy

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## Keywords

acute kidney injury; ischemic nephropathy; renal artery disease; renal failure

## Questions

1. A 70-kg man will have the greatest change in glomerular filtration rate (GFR) when the creatinine changes from:
    a. 0.9 to 1.2 mg/dL.
    b. 1.8 to 1.9 mg/dL.
    c. 3.2 to 3.9 mg/dL.
    d. 4.1 to 4.7 mg/dL.
    e. 7.9 to 11 mg/dL.
2. In patients with occult renal artery stenosis, angiotensin-converting enzyme (ACE) inhibitors cause acute renal failure (ARF) due to:
    a. sodium retention.
    b. increased antidiuretic hormone.
    c. afferent arteriolar vasoconstriction.
    d. efferent arteriolar vasodilation.
    e. decreased sympathetic nervous system activity.

3. Six days after partial nephrectomy in a solitary kidney, the patient is oliguric. Large amounts of fluid are coming from the flank drain. The serum creatinine increases from 1.7 to 3.2 mg/dL. The next step in management is:
    a. renal angiography.
    b. computed tomography (CT) scan with intravenous contrast.
    c. renal scan.
    d. immediate surgical exploration.
    e. magnetic resonance imaging (MRI).
4. After a 7-hour-long, complex urethral reconstruction performed in the extended lithotomy position, a patient has severe thigh and buttock pain. The creatinine phosphokinase (CPK) is dramatically elevated. The next step is:
    a. dopamine infusion.
    b. plasmapheresis.
    c. dobutamine infusion.
    d. forced alkaline diuresis.
    e. dialysis.
5. The sentinel cellular change in renal ischemic injury is:
    a. loss of cell polarity.
    b. depletion of adenosine triphosphate (ATP).
    c. alteration of $Na^+$ metabolism.
    d. increased intracellular $Ca^{2+}$.
    e. increased oxidant stress.
6. The renal structure at greatest risk for ischemic injury is the:
    a. afferent arteriole.
    b. cortical collecting duct.
    c. juxtaglomerular apparatus.
    d. straight segment (S3) proximal tubule.
    e. distal convoluted tubule.
7. A patient with acute kidney injury (AKI) has a urinary sodium of 10 mEq/L, urinary osmolality of 650, and a renal failure index of < 1. Urinalysis shows 10 to 20 red blood cells (RBCs) per high-power field (HPF), 3 to 5 white blood cells per HPF, 2 + proteinuria, and RBC casts. The most likely diagnosis is:

a. acute tubular necrosis (ATN).
b. prerenal azotemia.
c. acute glomerulonephritis.
d. acute interstitial nephritis.
e. obstruction.

8. When AKI is first recognized in a patient, the initial therapeutic intervention should be to:
   a. begin low-dose dopamine.
   b. administer a cardiac inotropic agent.
   c. restore adequate circulating blood volume.
   d. administer a loop diuretic.
   e. begin a mannitol infusion.
9. Loop diuretics are of benefit in the management of AKI due to:
   a. improved patient survival.
   b. decreased metabolic demand.
   c. decreased hypoxic cell swelling.
   d. free radical scavenging.
   e. increased renal vascular resistance.
10. The major risk of MRI with gadolinium in patients with advanced chronic kidney disease (CKD) is:
   a. nephrotoxicity.
   b. anaphylaxis.
   c. nephrogenic systemic fibrosis.
   d. seizures.
   e. hepatotoxicity.
11. Dopamine therapy in AKI:
   a. causes efferent arteriolar vasodilation.
   b. is recommended for routine use after renal transplantation.
   c. is effective due to improved cardiac function.
   d. is an unproven treatment.
   e. improves patient survival.
12. A patient with AKI after partial nephrectomy has a serum potassium of 6.9 mEq/L and widening of the QRS complex on electrocardiogram (ECG). The initial step in management should be:
   a. intravenous (IV) calcium.

b. IV insulin and glucose.
c. sodium polystyrene sulfonate resin (Kayexalate).
d. IV furosemide.
e. dialysis.

13. A patient with a serum creatinine level of 2.7 mg/dL requires renal angiography. The best way to protect renal function is:
    a. saline diuresis.
    b. pre-study mannitol.
    c. furosemide before study.
    d. dopamine throughout the study.
    e. atrial natriuretic factor before study.
14. In response to a reduction in renal mass, a number of events occur within the kidney that include all of the following except:
    a. activation of the sympathetic nervous system.
    b. hyperfiltration.
    c. glomerular hypertrophy.
    d. intrarenal vascular occlusion.
    e. interstitial fibrosis.
15. A 65-year-old man has a radical nephrectomy. The estimated GFR by the Modification of Diet in Renal Disease (MDRD) equation is 52 mL/min. Follow-up should include:
    a. low-protein diet.
    b. renal transplant evaluation.
    c. nephrology consult for stage 3 CKD.
    d. reassessment of kidney function every few months.
    e. loop diuretics.
16. A hypertensive patient with CKD should take an ACE inhibitor drug to:
    a. improve renal function.
    b. prevent progressive kidney disease.
    c. improve cardiac ejection fraction.
    d. enhance glycemic control.
    e. control blood lipids.
17. The most common cause for end-stage renal disease (ESRD) in the United States is:
    a. focal segmental glomerulosclerosis (FSGS).

b. membranoproliferative glomerulonephritis (type 2).
c. membranous glomerulonephritis.
d. autosomal dominant polycystic kidney disease.
e. diabetes mellitus.

18. The patient at lowest risk for progressive CKD is:
    a. diabetic, GFR = 86, albuminuria >300.
    b. postnephrectomy, GFR = 62 mL/min, albuminuria ≤30 mg/g.
    c. hypertensive, GFR = 75 mL/min, albuminuria = 80 mg/g.
    d. IgA nephropathy, GFR 42 mL/min, albuminuria = 70 mg/g.
    e. autosomal dominant polycystic kidney disease (ADPKD), GFR 28 mL/min, albuminuria ≤ 30 mg/g.
19. A hypertensive 38-year-old man has a serum creatinine of 2.4 mg/dL. The urinalysis has 10 to 20 RBCs/HPF, 3 + protein, and RBC casts. Ultrasound shows echogenic kidneys without hydronephrosis. The best way to achieve a diagnosis is:
    a. renal angiography.
    b. renal biopsy.
    c. retrograde pyelography.
    d. MRI.
    e. spiral CT scan.
20. All of the following promote fibrosis in the kidney except:
    a. angiotensin II.
    b. aldosterone.
    c. atrial natriuretic peptide.
    d. transforming growth factor–β.
    e. high-salt diet.
21. CKD patients treated with an ACE inhibitor may experience a decrease in residual renal function in the setting of:
    a. unilateral renal artery stenosis.
    b. concomitant treatment with an alpha-blocker.
    c. acquired renal cystic disease.
    d. left ventricular hypertrophy.
    e. ADPKD with cysts >10 cm.
22. The best renal replacement therapy for an otherwise healthy 37-year-old woman with chronic interstitial nephritis is:
    a. preemptive transplantation.

b. stabilize with hemodialysis 1 year, then transplant.
c. stabilize with peritoneal dialysis 1 year, then transplant.
d. home hemodialysis.
e. peritoneal dialysis with an automated cycler.

23. Hospitalization in ESRD patients on hemodialysis is most commonly due to:
    a. hypertension.
    b. ileus.
    c. diabetes.
    d. hyperkalemia.
    e. access catheter infection.
24. All the following have a direct toxic effect on the kidney except:
    a. iodinated contrast agent.
    b. myoglobin.
    c. gadolinium-based contrast agents.
    d. carboplatin.
    e. aminoglycoside antibiotics.
25. The strongest predictor of hospitalization in chronic dialysis patients is:
    a. African American race.
    b. hematocrit <30%.
    c. glomerulonephritis.
    d. poor nutritional status.
    e. age <30 years.

## Imaging

1. See Fig. 87.1. A 55-year-old woman had this abdominal radiograph 1 day after a contrast-enhanced CT scan was done for abdominal pain. Her creatinine before the CT scan was 1.9 mg/dL. The most likely diagnosis is:
    a. normal film.
    b. ATN.
    c. renal artery occlusion.
    d. hypertensive kidneys.

e. nephrocalcinosis.

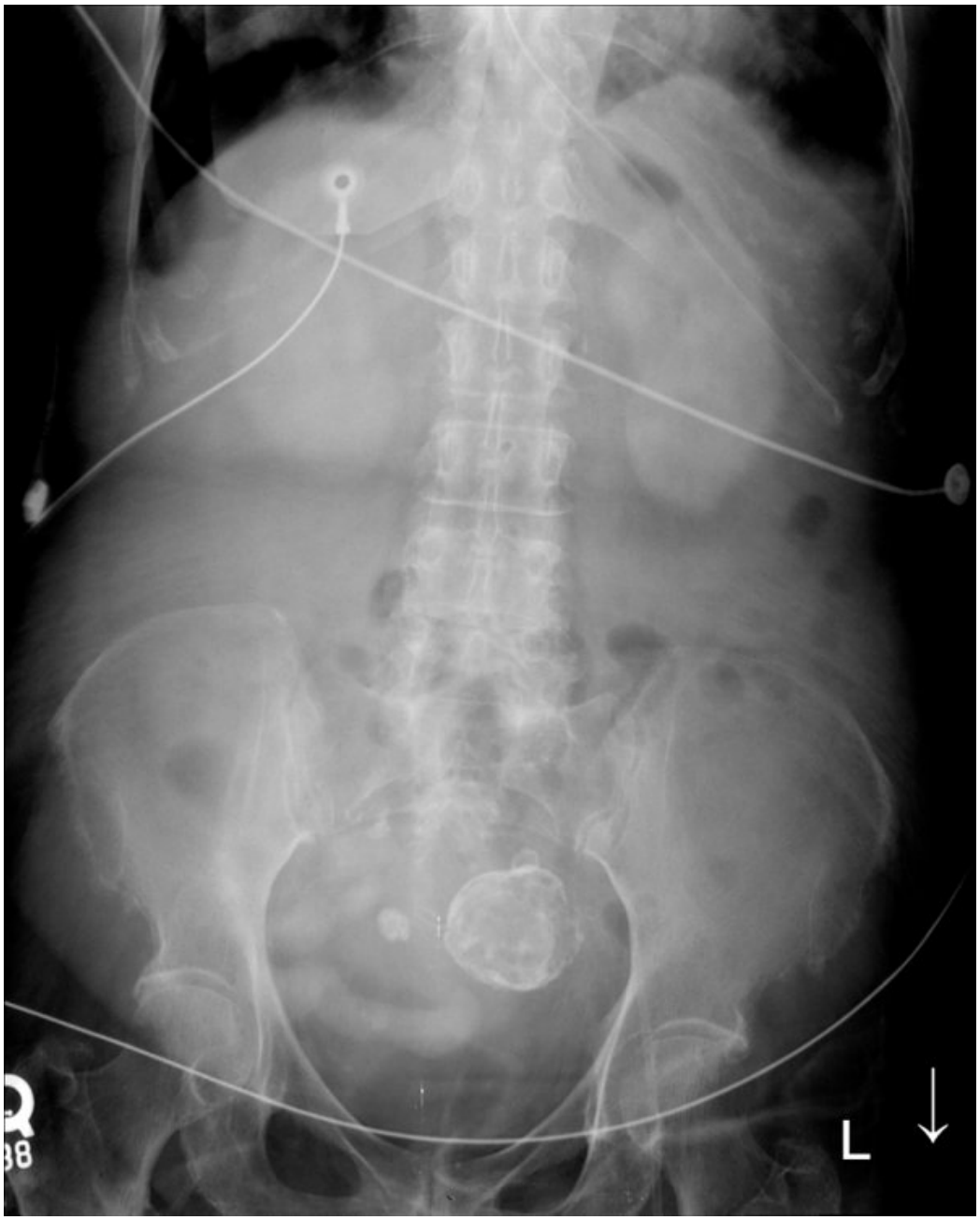



FIG. 87.1

## Answers

1. **a. 0.9 to 1.2 mg/dL.** The relationship between serum creatinine is not linear. **Above a GFR of 60 mL/min, changes in serum creatinine are minimal.** The change in serum creatinine from 0.9 to 1.2 may be associated with a large drop in GFR (120 to 60 mL/min). Below a GFR of 60, proportional increments in creatinine are associated with smaller changes in GFR. This is most notable at very high creatinine levels.
2. **d. Efferent arteriolar vasodilatation.** Angiotensin II has selectively greater vasoconstrictor effects on the efferent than on the afferent arteriole, whereas vasodilatory prostaglandins cause afferent arteriolar vasodilatation. Drugs that block angiotensin II synthesis (ACE inhibitors), block angiotensin II receptor binding (angiotensin II receptor antagonists), or inhibit vasodilatory prostaglandin synthesis (nonsteroidal anti-inflammatory drugs) may cause ARF in selected clinical settings.
3. **c. Renal scan.** There are several ways to confirm urinary extravasation. These include assessment of drain fluid for creatinine, intravenous administration of a vital dye excreted by the kidneys (such as indigo carmine or methylene blue), and radiographic demonstration of a fistula (isotope renography, retrograde pyelography, cystography, computed tomography). **Renal scan can assess perfusion and also demonstrate extravasation.**
4. **d. Forced alkaline diuresis. The combination of renal hypoperfusion and the nephrotoxic insult of myoglobin or hemoglobin within the proximal tubule may result in ATN. Early recognition of this disorder is crucial, because a forced alkaline diuresis is indicated to minimize nephrotoxicity.**
5. **b. Depletion of adenosine triphosphate (ATP).** The sentinel biochemical event in renal ischemia is the depletion of ATP, which is the major energy currency for cellular work.
6. **d. Straight segment (S3) proximal tubule.** The S3 segment of the proximal tubule is associated with the greatest ischemic damage. Other structures that sustain injury in this region include the

medullary thick ascending limb, which is metabolically active and rich in the energy-requiring $Na^+$, $K^+$-ATPase.

7. **c. Acute glomerulonephritis. A low fractional excretion of sodium (or renal failure index) may be associated with either prerenal azotemia or acute glomerulonephritis.** These entities could be separated clinically by examination of the urinalysis results. Conditions associated with prerenal azotemia would have a bland urinalysis, whereas proteinuria, RBCs, and RBC casts would be seen with acute glomerulonephritis.
8. **c. Restore adequate circulating blood volume.** During the initial stages, a trial of parenteral hydration with isotonic fluids may correct ARF secondary to prerenal causes.
9. **b. Decreased metabolic demand.** Loop diuretics decrease active NaCl transport in the thick ascending limb of Henle and thereby limit energy requirements in the metabolically active segment, which often bears the greatest ischemic insult.
10. **c. Nephrogenic systemic fibrosis. Recently, gadolinium-based contrast agents have been associated with the development of nephrogenic systemic fibrosis.** At-risk patients include those with advanced CKD. It is important that these compounds be given to these patients only after careful consideration of the indication for the study.
11. **d. Is an unproven treatment.** Results of clinical studies have not conclusively proved that dopamine infusion improves ARF.
12. **a. Intravenous (IV) calcium. Priorities for treatment of acute hyperkalemia with electrocardiographic changes include stabilizing the electrical membrane of the cardiac conduction system, which may be accomplished with the use of intravenous calcium salts. These have an immediate effect and a rather short duration of action.**
13. **a. Saline diuresis.** A study by Solomon and coworkers confirmed that prestudy intravenous hydration with saline was crucial in limiting the nephrotoxic effect of radiocontrast agents in patients with preexisting azotemia. The addition of either a loop diuretic or mannitol did not improve outcome.

14. **d. Intrarenal vascular occlusion.** In response to reduced nephron mass, a mosaic of events occurs linking sympathetic nervous system activation, renal structural remodeling, altered gene expression and regulation, and several regulatory mechanisms for progression.
15. **c. Nephrology consult for stage 3 CKD.** According to the National Kidney Foundation (K/DOQI) guidelines, this patient indeed has stage 3 CKD. A nephrologist should be following this patient and appropriate preventive strategies instituted for preservation of kidney function and minimizing the impact of comorbidities.
16. **b. Prevent progressive kidney disease. ACE inhibitors work by hemodynamic and nonhemodynamic mechanisms to slow the progression of renal disease.**
17. **e. Diabetes mellitus. Diabetes mellitus and hypertension account for the greatest percentage of cases, followed by glomerular diseases (e.g., FSGS, membranous glomerulonephritis), and then secondary glomerulonephritis associated with systemic diseases (e.g., systemic lupus erythematosus, Wegener granulomatosis).**
18. **b. Postnephrectomy, GFR = 62 mL/min, albuminuria $\leq$ 30 mg/g.** The risk of progressive renal disease is based on the cause for renal function reduction, the GFR, and the albuminuria score. In a and c the GFR is higher, and there is cause for ongoing injury. One-time reduction in GFR with nephrectomy in the absence of ongoing injury (no albuminuria) should yield a more favorable prognosis.
19. **b. Renal biopsy.** For definitive diagnosis, a renal biopsy is required to aid prognosis and therapy decisions, especially in the setting of abnormal renal function.
20. **c. Atrial natriuretic peptide.** All choices promote renal fibrosis except atrial natriuretic peptide, which may have the opposite effect.
21. **e. ADPKD with cysts larger than 10 cm. Individuals with bilateral renal artery stenosis and autosomal dominant PKD patients with cyst size greater than 10 cm may also experience a**

**decrease in residual renal function while being given ACE inhibitor therapy.**

22. **a. Preemptive transplantation.** A comparison of outcomes suggests that renal transplantation is the best overall treatment for ESRD patients.
23. **e. Access catheter infection.** The most recent USRDS data identify the most common reason for hospital admission in ESRD patients on hemodialysis is catheter infection.
24. **c. Gadolinium-based contrast agents.** Gadolinium is not toxic to the kidney. Its risk relates to the development of nephrogenic systemic fibrosis, which can carry a high mortality in patients with severely reduced GFR. All the others demonstrate direct toxicity to the kidney.
25. **d. Poor nutritional status.** The strongest predictors of the number of hospitalizations per year of patients at risk include low serum albumin, decreased activity level, diabetes mellitus as a primary cause of ESRD, peripheral vascular disease, white race, increasing age, and congestive heart failure. Both nutritional status (levels of serum albumin, creatinine, transferrin, and prealbumin, and lean body mass) and inflammatory response (e.g., C-reactive protein) are independent predictors of hospitalization in chronic hemodialysis patients.

## Imaging

1. **b. ATN.** Both kidneys are diffusely dense 24 hours after contrast administration (persistent nephrogram), with no excretion into the collecting system, making ATN the best diagnosis. The process is likely related to contrast-induced nephropathy. Renal artery occlusion is usually a unilateral process, and the nephrogram is absent in the affected kidney because of the lack of perfusion. The kidneys are small in patients with severe hypertension, and the nephrogram would have washed out at 24 hours rather than be persistent. No calcifications are seen in the kidneys, making nephrocalcinosis an unlikely diagnosis.

## Chapter Review

1. ARF is divided into prerenal, intrarenal, and postrenal.
2. Prerenal failure is the result of decreased renal blood flow or increased nitrogen load. The serum blood urea nitrogen-to-creatinine ratio is greater than 10:1. Urine volume is generally low, osmolality high, and sodium content very low.
3. Intrarenal failure is due to parenchymal disease. It may involve the glomerulus as in acute glomerulonephritis, the tubule as in ATN, or the interstitium as in acute interstitial nephritis. The serum blood urea nitrogen-to-creatinine ratio is 10:1. Urine output is variable, and urinary sodium is generally above 20 mEq/L. If the cause is ATN, urinary sodium is generally greater than 40 mEq/L.
4. Postrenal failure is due to obstruction of the entire nephron mass. Patients who are anuric should be suspected of having complete urinary tract obstruction, acute cortical necrosis, or bilateral vascular occlusion. The serum blood urea nitrogen-to-creatinine ratio is 10:1.
5. In ATN, renal blood flow is reduced by 50% or more with the perfusion defect most marked in the outer medulla. Tubule permeability is also increased.
6. The use of mannitol before an ischemic insult has been shown to be of benefit.
7. Hyperkalemia, as evidenced on the electrocardiogram by peaked T-waves, a prolonged PR interval, and widening of the QRS complex is initially treated with IV calcium salts to stabilize the myocardium and then transiently lower serum potassium with IV glucose and insulin or IV sodium bicarbonate, as well as to permanently lower the potassium with either Kayexalate or dialysis. Administration of Kayexalate has been associated with colonic necrosis
8. The indications for initiation of dialysis include volume overload, severe hyperkalemia, severe metabolic acidosis, pericarditis, selected poisonings, and uremic symptomatology.
9. Peritoneal dialysis is less stressful hemodynamically.
10. The mortality rate for patients with ATN approximates 50%. Of those who survive ATN, 5% will require chronic dialysis.

11. Patients who are to receive IV contrast and are at risk for renal failure should be given IV hydration with saline before the use of nonionic IV contrast.
12. CKD is defined as a GFR less than 60 mL/min.
13. Hyperfiltration results when a decreased number of nephrons are called on to perform the entire workload. This results in elevated glomerular hydrostatic pressure, which is a major contributor to decreased renal function.
14. A family history of ESRD is a strong predictor of future risk for renal failure.
15. The Cockcroft Gault formula, MDRD equation, and CKD-EPI equation are all used to estimate GFR from serum creatinine. These equations are inaccurate for patients who have changing renal function, are at the extremes of age or body size, are obese, have decreased muscle mass, or are sick with moderately advanced renal failure.
16. Cystatin-C, a serum protein, may be more predictive of renal function than creatinine.
17. Mortality rates secondary to sepsis are one to several hundred times higher in dialysis patients.
18. Creatinine is secreted by the renal tubule and may overestimate renal function, particularly at low GFRs. Cimetidine and trimethoprim block this secretion.
19. Drugs that block angiotensin II production or action appear to decrease the rate of decline in renal function in patients with chronic renal disease.
20. In urology, rhabdomyolysis is associated with the protracted exaggerated lithotomy and lateral decubitus positions.
21. The sentinel event in renal ischemia is depletion of ATP.
22. The presence of albumen in the urine of patients with chronic renal failure is an important predictor of the rate of decline in renal function.
23. When the GFR falls below 50% of normal, progressive loss of renal function occurs even though the cause of the renal failure is inactive or no longer present.

24. The remaining lifetime for patients on dialysis is 15% to 25% of the general population.
25. Myoglobin can be nephrotoxic to the proximal tubule and may result in ATN. Early recognition of muscle necrosis with resultant myoglobinuria is crucial because a forced alkaline diuresis is indicated to minimize nephrotoxicity.
26. A low fractional excretion of sodium (or renal failure index) may be associated with either prerenal azotemia or acute glomerulonephritis.
27. Gadolinium-based contrast agents have been associated with the development of nephrogenic systemic fibrosis. Patients at risk are those with advanced CKD.
28. Individuals with bilateral renal artery stenosis and those with autosomal dominant PKD with cyst size greater than 10 cm may experience a decrease in residual renal function when receiving ACE inhibitor therapy.

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# Urological Complications of Renal Transplantation

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## Keywords

kidney transplant; renal transplantation; urological complications

## Questions

1. In renal transplant recipients, all of the following are true EXCEPT:
    a. Recipients usually have classical symptoms of ureteral obstruction.
    b. Ureteral stricture is the most common cause of late ureteral obstruction.
    c. Ureteral obstruction should be suspected in any deterioration of renal function.
    d. Ureter ischemia is the main etiology for ureteral stricture formation.
2. The predominant type of stones in renal transplant patients is:
    a. calcium oxalate.
    b. calcium phosphate.
    c. struvite.
    d. uric acid.
    e. a and b.
3. Early stent removal is associated with:
    a. decrease in incidence of urinary tract infection (UTI).

b. decrease in risk of stent colonization.
c. increase in risk of UTI.
d. increase in risk of developing urine leak.
e. increase in risk of graft rejection.

4. All of the following have been related to the development of clinically significant vesicoureteral reflux (VUR) in renal transplant recipients EXCEPT:
   a. bladder outlet obstruction.
   b. neurogenic detrusor overactivity.
   c. reduced bladder capacity and compliance.
   d. ureterovesical anastomosis technique.
   e. use of ureteral stent.
5. The success rate of laparoscopic marsupialization of lymphocele in renal transplant recipients is:
   a. 30%.
   b. 41%.
   c. 69%.
   d. 92%.
   e. 84%.
6. The following statements regarding renal cell carcinoma in renal transplant recipients are true EXCEPT:
   a. Most of the kidney tumors develop in the native kidneys.
   b. Renal cell carcinoma incidence is higher in the transplant population compared to the general population.
   c. Renal cell carcinoma in renal transplant patients is aggressive and associated with worse oncological outcome.
   d. Partial nephrectomy is a feasible option to treat small renal masses in the allograft.
   e. Mass screening for renal cell carcinoma in renal transplant recipients is not recommended.
7. The following statements regarding the management of benign prostatic hypertrophy (BPH) in renal transplant recipients are FALSE EXCEPT:
   a. Transurethral resection of prostate (TURP) can be done in the first 3 weeks after transplant.
   b. TURP can be done in the presence of ureteral stent.

c. TURP should be avoided in the first 3 weeks after transplant.
d. Medical treatment of BPH should be avoided in renal transplant recipients.
e. Renal transplant recipients are at increased risk of developing urethral stricture after TURP.

8. The following statements regarding the management of bladder cancer in renal transplant recipients are true EXCEPT:
   a. Intravesical bacillus Calmette-Guerin can be used in renal transplant patients.
   b. Muscle-invasive bladder cancer can be treated by chemoradiotherapy.
   c. Muscle-invasive bladder cancer can be treated by radical cystectomy and urinary diversion.
   d. Early cystectomy should be offered to patients with high-grade non–muscle-invasive bladder cancer.
   e. Bladder cancer in renal transplant recipients has indolent behavior.
9. All of the following are predisposing factors to the development of lymphocele EXCEPT:
   a. Extensive dissection of the lymphatic around the iliac vessels.
   b. Use of mammlian target of rapamycin (mTOR) inhibitors.
   c. Prolonged warm ischemia time.
   d. Delayed graft function.
   e. Atherosclerosis.
10. Initial management of a low-volume urine leak may include all of following EXCEPT:
   a. bladder catheter.
   b. JJ stent placement.
   c. percutaneous nephrostomy tube placement.
   d. percutaneous drainage of urinoma.
   e. fluid restriction.
11. Which of the following is a contraindication for surgical reconstruction of ureteral stricture?
   a. Strictures less than 3 cm in length.
   b. Late stricture recurrence after endourological treatment.
   c. Surgeon preference.

d. All of above.
e. None of above.

## Answers

1. **a. Recipients usually have classical symptoms of ureteral obstruction.** Due to renal allograft during the procurement process, kidney transplant recipients rarely manifest the classical symptoms of ureteral obstruction. As a result, usually recipients present with a progressive decline in renal function, a decrease in urine output, and recurrent UTIs. In a rare instance, patients may present with poorly characterized ache over the allograft due to irritation of peritoneum.
2. **e. a and b.** The most common stone composition in recipients is calcium-based stones (67%), followed by struvite stones (20%) and uric acid stones (13%).
3. **a. Decrease in incidence of urinary tract infection (UTI).** In a recent meta-analysis, early stent removal, defined by removal of the stent before the third postoperative week (< day 15), has been shown to reduce the incidence of UTI without increasing the incidence of urine leak and stenosis. On the other hand, 27% of the ureteral stents are colonized regardless of the indwelling time.
4. **d. Ureterovesical anastomosis technique.** VUR can be related to a technical error during the ureteroneocystostomy anastomosis, bladder outlet obstruction, or high-pressure urine storage secondary to either detrusor overactivity or reduced bladder compliance. However, VUR can develop regardless the ureterovesical anastomosis technique used.
5. **d. 92%.** The success rate for aspiration alone, sclerotherapy using different composite, drain placement, laparoscopic, and open surgery is 41%, 69%, 50%, 92%, and 84%, respectively.
6. **c. Renal cell carcinoma in renal transplant patients is aggressive and associated with worse oncological outcome.** Majority of kidney graft tumors has indolent behavior. In a recent analysis of allograft tumors, it was found that papillary carcinomas are

reported in 56% of kidney graft tumors and low-grade tumors (Fuhrman grades 1 and 2) accounted for 65% of nonpapillary kidney graft tumors.

7. **c. TURP should be avoided in the first 3 weeks after transplant.** There is no contraindication for medical therapy in renal transplant recipients. TURP or Holmium enucleation of the prostate has been reported to be safe and effective in this subset of patients if performed at least 3 weeks post-transplant. It is crucial to avoid TURP in the first 2 weeks and in the presence of a ureteral stent after the transplant as it has been associated with devastating complications such as sepsis and death.
8. **e. Bladder cancer in renal transplant recipients has indolent behavior.** Bladder cancer in kidney transplant patients is aggressive and tends to be associated with higher recurrence rates, progression, and metastasis. For that reason, early cystectomy should be offered to patients with high-grade non–muscle-invasive bladder cancer. However, intravesical bacillus Calmette-Guerin can be used in selected patients with high-risk non–muscle-invasive bladder cancer. Muscle-invasive bladder cancer can be treated with chemoradiotherapy or radical cystoprostatectomy and urinary diversion.
9. **e. Atherosclerosis.** Lymphoceles occur mainly due to extensive dissection of the lymphatics around the iliac vessels of the recipient or renal vessels of the donor occurring during the time of organ procurement surgery or back table preparation. Other factors found to be associated with lymphocele formation include obesity, recipient age, duration of dialysis treatment, warm ischemia time, use of prophylaxis low-molecular-weight heparin, delayed graft function, acute rejection, redo transplant, and the use of mTOR inhibitors.
10. **e. Fluid restriction.** The initial treatment of urine leak is by diverting the urine from the leak site, and this can be achieved by a Foley catheter and a percutaneous nephrostomy tube. Retrograde stent placement has also been proposed when a double-J stent was not placed at the time of surgery. Percutaneous drainage is

recommended if a substantial urinoma forms or causes ureteral obstruction or there is evidence of infection.

11. **e. None of above.** Open surgical repair for ureteral stricture remains the most effective treatment modality with superior long-term results when compared to endoscopic management. The only relative contraindication is very highly co-morbid patients with short life expectancy.

## Chapter Review

1. Majority of postoperative hematuria is self-limiting.
2. Consider duplex US for severe or persistent hematuria after renal biopsy to rule out arteriovenous (AV) fistula.
3. Ureteral stent decreases ureteral stricture and urinary leak.
4. Individualized stent removal can be after 3 weeks post-transplantation.
5. Strictures are the predominant cause of ureteral obstruction in patients greater than 3 months post-transplantation.
6. The primary etiology of ureteral stricture development is ischemia.
7. Stricture should be suspected in patients with hydronephrosis and decreased graft function.
8. A variety of endourological treatment options have been described to treat ureteral strictures, however, open reconstructive surgery offers superior long-term results.
9. Most of the kidney tumors develop in the native kidneys and are low-grade tumors.
10. Active surveillance, radiation, and surgery (open and minimally invasive) are reasonable options to treat prostate cancer in kidney transplant patients.
11. Kidney transplant recipients have a threefold risk of urothelial carcinoma (UC) compared with the general population, and it is biologically more aggressive.